Lab Exercise 6- Docker-Compose file

Objective:

Set up a WordPress environment using Docker Compose, including a MySQL database as the backend.

Prerequisites:

- Docker and Docker Compose installed on your system.

Step 1: Create a docker-compose.yml File

1. In the project directory, create a file named docker-compose.yml.
2. Add the following content to docker-compose.yml:

docker-compose.yml

```
version: '3.8'

services:
  wordpress:
    image: wordpress:latest
    ports:
      - "8002:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wp_user
      WORDPRESS_DB_PASSWORD: wp_pass
      WORDPRESS_DB_NAME: wp_database
    depends_on:
      - db

  db:
    image: mysql:latest
    environment:
      MYSQL_ROOT_PASSWORD: root_password
      MYSQL_DATABASE: wp_database
```

```
      MYSQL_USER: wp_user
      MYSQL_PASSWORD: wp_pass
    volumes:
      - db_data:/var/lib/mysql

volumes:
  db_data:
```

Step 2: Start the Containers

1. Run the following command to start the containers:

```
docker-compose up -d
```

2. Docker Compose will download the necessary images (WordPress and MySQL) and start both services.

Step 4: Access WordPress

1. Open your web browser and go to **http://localhost:8002**
2. Follow the WordPress installation steps to set up your site.

Step 5: Stop and Remove Containers

To stop the containers and remove the associated resources, run:

```
docker-compose down
```

Explanation of docker-compose.yml:

- **wordpress**: Sets up the WordPress container, mapping port 80 inside the container to port 8002 on your local machine.
- **db**: Sets up the MySQL container with a volume (db_data) for persistent storage.

Additional Notes:

- Modify the environment variables as needed for different configurations.

- To view logs, use docker-compose logs -f.

This setup allows you to quickly start a WordPress site locally and experiment with configurations.

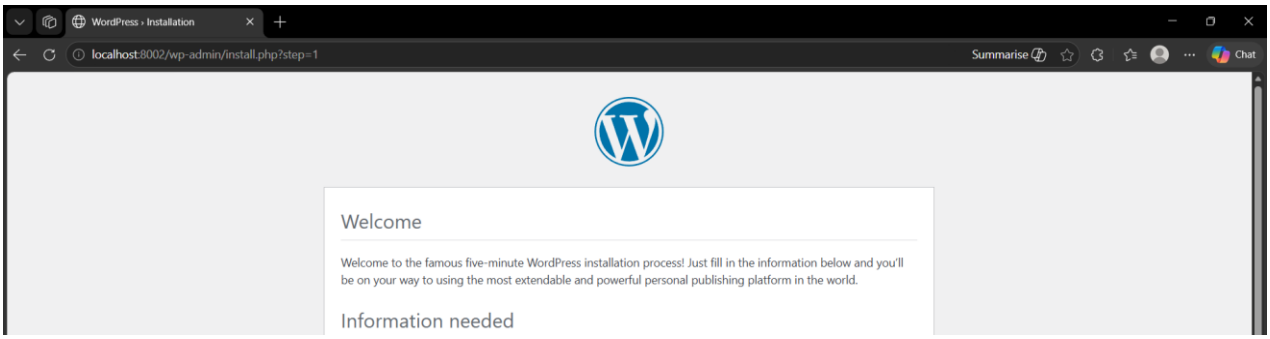
```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>mkdir wordpress-docker

C:\Users\NK>cd wordpress-docker

C:\Users\NK\wordpress-docker>(echo version: '3.8' & echo. & echo services: & echo   wordpress: & echo     image: wordpre
ss:latest & echo     ports: & echo       - "8002:80" & echo     environment: & echo       WORDPRESS_DB_HOST: db:3306 & e
cho       WORDPRESS_DB_USER: wp_user & echo       WORDPRESS_DB_PASSWORD: wp_pass & echo       WORDPRESS_DB_NAME: wp_data
base & echo     depends_on: & echo       - db & echo. & echo   db: & echo     image: mysql:latest & echo     environment
: & echo       MYSQL_ROOT_PASSWORD: root_password & echo       MYSQL_DATABASE: wp_database & echo       MYSQL_USER: wp_u
ser & echo       MYSQL_PASSWORD: wp_pass & echo     volumes: & echo       - db_data:/var/lib/mysql & echo. & echo volume
s: & echo   db_data:) > docker-compose.yml

C:\Users\NK\wordpress-docker>docker-compose up -d
time="2026-04-28T08:30:15+05:30" level=warning msg="C:\\Users\\NK\\wordpress-docker\\docker-compose.yml: the attribute `
version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 40/40
[+] up 44/44ql:latest     Pulled                                                                                    35.1s
 ✔Image mysql:latest                    Pulled                                                                     35.1s
 ✔Image wordpress:latest                Pulled                                                                     31.3s
 ✔Network wordpress-docker_default      Created                                                                     0.1s
 ✔Volume wordpress-docker_db_data       Created                                                                     0.0s
 ✔Container wordpress-docker-db-1       Created                                                                     0.6s
 ✔Container wordpress-docker-wordpress-1 Created                                                                    0.2s
```



```
C:\Users\NK\wordpress-docker>docker-compose ps
time="2026-04-28T08:31:13+05:30" level=warning msg="C:\\Users\\NK\\wordpress-docker\\docker-compose.yml: the attribute `version` is obsolete, it will be ign
ored, please remove it to avoid potential confusion"
NAME                          IMAGE              COMMAND                  SERVICE     CREATED          STATUS          PORTS
wordpress-docker-db-1         mysql:latest       "docker-entrypoint.s…"   db          22 seconds ago   Up 21 seconds   3306/tcp, 33060/tcp
wordpress-docker-wordpress-1  wordpress:latest   "docker-entrypoint.s…"   wordpress   22 seconds ago   Up 21 seconds   0.0.0.0:8002->80/tcp, [::]:8002->80/
tcp

C:\Users\NK\wordpress-docker>http://localhost:8002
'http:' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK\wordpress-docker>docker-compose logs -f
time="2026-04-28T08:32:11+05:30" level=warning msg="C:\\Users\\NK\\wordpress-docker\\docker-compose.yml: the attribute `version` is obsolete, it will be ign
ored, please remove it to avoid potential confusion"
wordpress-1  | WordPress not found in /var/www/html - copying now...
wordpress-1  | Complete! WordPress has been successfully copied to /var/www/html
wordpress-1  | No 'wp-config.php' found in /var/www/html, but 'WORDPRESS_...' variables supplied; copying 'wp-config-docker.php' (WORDPRESS_DB_HOST WORDPRES
S_DB_NAME WORDPRESS_DB_PASSWORD WORDPRESS_DB_USER)
wordpress-1  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'ServerName' directive glo
bally to suppress this message
wordpress-1  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'ServerName' directive glo
bally to suppress this message
wordpress-1  | [Tue Apr 28 03:00:53.138713 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.66 (Debian) PHP/8.3.30 configured -- resuming normal
 operations
db-1         | 2026-04-28 03:00:52+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.7.0-1.el9 started.
wordpress-1  | [Tue Apr 28 03:00:53.138838 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
db-1         | 2026-04-28 03:00:52+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
wordpress-1  | 172.18.0.1 - - [28/Apr/2026:03:01:25 +0000] "GET / HTTP/1.1" 302 424 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0"
db-1         | 2026-04-28 03:00:52+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.7.0-1.el9 started.
db-1         | 2026-04-28 03:00:52+00:00 [Note] [Entrypoint]: Initializing database files
db-1         | 2026-04-28T03:00:52.949161Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
db-1         | 2026-04-28T03:00:52.950985Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 9.7.0) initializing of server in progress as process 80
db-1         | 2026-04-28T03:00:52.968183Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
db-1         | 2026-04-28T03:00:53.540314Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
db-1         | 2026-04-28T03:00:55.717463Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off
 the --initialize-insecure option.
```